

ZAKIYA WILLIAMS

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EDUCATION

M.S., Engineering Management (Specialty: Cybersecurity)
Johns Hopkins University, Whiting School of Engineering

Starting Spring 2026

M.S., Electrical and Computer Engineering
North Carolina A&T State University

December 2025

PROFESSIONAL SUMMARY

Relentless problem solver with an M.S. in Electrical & Computer Engineering and practical experience detecting, triaging, and analyzing security events across Windows and Linux environments. Experienced in simulating endpoint attacks mapped to MITRE ATT&CK, tuning Wazuh detection rules, and improving SOC processes. Backed by Security+ and CCNA certifications, strong technical communication, and leadership experience across high pressure academic and field roles.

TECHNICAL SKILLS

Security & SIEM	Wazuh (manager, indexer, dashboard), alert rules & decoders, log analysis, MITRE ATT&CK aligned adversary emulation
Networking	CP/IP, VLANs, routing basics, DHCP, NAT, VPN concepts, network monitoring, troubleshooting
Systems & Virtualization	Linux (Ubuntu, Debian), Windows 10/11, KVM/QEMU, Proxmox, VMware, server/endpoint hardening basics
Programming & Scripting	Python (data analysis, log parsing), Bash, $\text{\LaTeX}$; familiarity with YAML for automation scenarios
Data & Tools	Jupyter Notebooks, Pandas, CSV/JSON log pipelines, GitHub, basic SQL

CERTIFICATIONS

Cisco CCNA	December 2024
CompTia Security +	September 2024

PROJECTS

Graduate Researcher – Wazuh SIEM/XDR Evaluation <i>Master's Thesis</i>	2024-2025
- Designed and implemented a multi VM security lab using Ubuntu (KVM/QEMU), Windows 10/11, Debian, and Wazuh (manager, indexer, dashboard) to evaluate SIEM/XDR capabilities. - Used CALDERA and scripted adversary emulation (MITRE ATT&CK techniques T1059, T1078, T1082, T1053, T1046) to generate realistic endpoint and network telemetry. - Collected and analyzed alert data (CSV/JSON) in Python/Jupyter to measure detection accuracy, timeliness, rule diversity, and cross platform consistency for Windows vs. Linux endpoints. - Used python scripts to create visualizations (heatmaps, bar charts, timelines) to communicate findings to faculty, security engineers, and non-technical audiences.	
Security Home Lab & GitHub Portfolio (Ongoing) GitHub: github.com/ZakiyaWilliams617	2025 - Present
- Building a home lab combining virtual and physical appliances to practice SIEM tuning, log pipelines, alert triage, and incident response playbooks. - Maintaining a GitHub portfolio with Jupyter notebooks, $\text{\LaTeX}$ reports, and configuration examples that demonstrate security engineering workflows and tooling familiarity.	

RESEARCH PUBLICATION

Williams, Z "Evaluating the Detection and Response Effectiveness of Wazuh Across Multiple Endpoint Threat Categories
(In preparation, 2025)

PROFESSIONAL EXPERIENCE

Independent Contractor

July 2023 - Present

Field Nation

- Network infrastructure solutions, specializing in the installation and security configurations for enterprise and retail clients.
- Performed comprehensive security implementations, including firewall configuration, router/switch setups, and network optimization for clients in diverse sectors.
- Diagnosed and resolved complex LAN/WAN vulnerabilities, enhancing the security and efficiency of client networks across multiple sites.
- Collaborated with different teams to ensure compliance with industry standards and mitigate potential security risks.

Network Infrastructure & Assistant Principal

July 2020 - June 2023

KIPP Chicago Schools

- Led the redesign and expansion of the school's network infrastructure, ensuring secure, stable connectivity across multiple campuses and endpoint devices.
- Introduced and developed a district wide Network Engineering program with built in modules on cybersecurity fundamentals, log analysis, and incident detection.
- Collaborated with different teams to implement protocols for system access, secure data handling, and incident response across staff and students.
- Coordinated deployment of endpoint security tools and helped shape the rollout of access controls and firewall rules across multiple VLANs.
- Documented technical procedures and trained non technical staff, effectively bridging the communication gap between tech teams and leadership.

Cybersecurity & Data Protection Analyst

September 2018 - June 2020

First Tech Challenge

- Managed network traffic flow and security configurations, implementing VPNs, encryption protocols, and endpoint protection tools to reduce exposure to threats.
- Responded to suspected intrusions by isolating endpoints, analyzing logs, and escalating incidents as needed and helping reduce false positives across student based networks.
- Applied patch management best practices and monitored systems for vulnerabilities using tools like antivirus platforms and manual log review.
- Trained team members on security essentials, basic IAM concepts, and best practices around data privacy, contributing to long term behavioral changes across users.
- Helped coordinate DLP style policies within educational environments by restricting access to sensitive student data and monitoring file movement

Network Security Program Manager

July 2018 - June 2020

Project Lead the Way

- Designed and implemented a district wide network security program, preparing students for Cisco CCNA and cybersecurity certifications while enhancing their technical skills.
- Integrated network security tools, firewalls, and routing protocols into lab environments, enabling hands-on learning experiences for students.
- Led several teams to establish a robust network infrastructure, implementing security best practices and ensuring the program's sustainability.

Science Robotics Instructor

June 2013 - June 2020

KIPP Chicago & Other schools

- Directed the development and integration of Robotics and Network Engineering curriculum, which included networking and cybersecurity training.
- Led network expansion across several schools, enhancing wireless connectivity, server performance, and cloud integration to support academic and administrative operations.
- Introduced and led a district wide Network Engineering Program, preparing students for industry certifications like Cisco CCNA, while also spearheading regional robotics competitions.
- Provided frontline network security and troubleshooting support, ensuring robust protection against cyber threats, while collaborating with IT teams to address hardware failures, configure firewalls, and secure wireless networks.

Flight Attendant

November 2013 - Present

Jetblue Airways

- Managed inflight emergencies, ensuring compliance with safety regulations while maintaining calm and effective communication during high pressure situations.
- Worked collaboratively with crew members to guarantee passenger safety and comfort, applying strict adherence to company and regulatory policies.

PROFESSIONAL MEMBERSHIPS

National Society of Black Engineers (NSBE)	2023 - Present
Society of Women Engineers (SWE)	2024 - Present